

Module 1: Networking Today

CCNA 1: Introduction to Networks v7.0 — Identification Questionnaire

Name: _____

Date: _____

Course & Year: NETC311 / Year III

Score: _____ / 99

GENERAL INSTRUCTIONS: This questionnaire contains 99 identification-type items based thoroughly on Module 1: Networking Today. Read each statement carefully. Identify the correct term that completes the blank (____) by selecting the best answer among choices a, b, c, or d. Write the letter of your choice on the blank provided before each number or encircle the letter. A complete Answer Key and Study Reference is provided at the end of this document.

PART I: IDENTIFICATION QUESTIONNAIRE

____ 1. Modern computer networks enable communication and collaboration across a global scale, creating a world without ____.

- a.) boundaries
- b.) delays
- c.) hardware
- d.) protocols

____ 2. Advancements in networking technology have allowed people to interact and share ideas in online groups known as global ____.

- a.) segments
- b.) communities
- c.) domains
- d.) subnets

____ 3. The global networking platform that supports the way we work, live, play, and learn is referred to by Cisco as the ____ network.

- a.) human
- b.) analog
- c.) converged
- d.) physical

____ 4. Every computer connected to a network that participates directly in network communication is classified as a host or ____.

- a.) intermediary device
- b.) physical interface
- c.) network medium
- d.) end device

____ 5. Computers on a network that provide information, files, or services to other end devices are called ____.

- a.) repeaters
- b.) servers
- c.) clients
- d.) multiplexers

____ 6. Computers that send requests to servers to retrieve information such as web pages, email, or documents are called ____.

- a.) concentrators
- b.) gateways
- c.) clients
- d.) hosts of last resort

____ 7. A specialized server that runs email server software to manage, send, and store electronic messages is an ____.

- a.) email server
- b.) print server
- c.) proxy firewall
- d.) FTP server

____ 8. To retrieve and display web pages hosted on a web server, clients run client software called a web ____.

- a.) sniffer
- b.) browser

- c.) compiler d.) terminal emulator

_____ 9. A dedicated computer on a network that stores corporate and personal files for authorized user access is called a _____.

- a.) file server b.) DNS server
c.) DHCP relay d.) domain gateway

_____ 10. A network design where a single computer can act as both a client and a server at the same time is called a _____.

- a.) client-server network b.) peer-to-peer network
c.) mainframe network d.) centralized cloud network

_____ 11. An advantage of a peer-to-peer network configuration is that it is less complex, easy to set up, and has a _____.

- a.) guaranteed throughput b.) lower cost
c.) high level of security d.) centralized database

_____ 12. A primary disadvantage of implementing a peer-to-peer network is that it has no _____.

- a.)** operating system requirement
b.) centralized administration
c.) IP addressing capability
d.) copper cable support

_____ 13. Peer-to-peer networks are only recommended for very small environments because they are not _____.

- a.) scalable b.) routable
c.) functional d.) digital

_____ 14. Because of security and performance limitations, peer-to-peer networks are typically used only for simple tasks such as transferring files and sharing _____.

- a.) routing tables b.) encryption certificates
c.) printers d.) VLAN configurations

_____ 15. A network device where a message originates from or where it is received is known as an _____.

- a.) intermediary device b.) core switch
c.) end device d.) boundary router

_____ 16. A device that interconnects end devices and manages data as it flows across the network is called an _____.

- a.) intermediary network device b.) passive transmission line
c.) end-user terminal d.) endpoint device

_____ 17. Examples of intermediary network devices include switches, routers, firewalls, and _____.

- a.) VoIP smartphones b.) workstations
c.) wireless access points d.) network printers

_____ 18. One primary management function of an intermediary device as data travels across a network is to regenerate and _____.

- a.) retransmit data signals b.) modify packet payloads
c.) reboot end hosts d.) format hard drives

_____ 19. Intermediary devices manage data flow by maintaining information about what pathways exist through the _____.

- a.) system bus b.) network
c.) operating system kernel d.) hard drive partition

_____ 20. When a network link failure or communication disruption occurs, an intermediary device notifies other devices of errors and communication _____.

- a.) physical topology diagram b.) conceptual data schema
c.) wiring flowchart d.) logical topology diagram

____ 32. A network diagram that illustrates device names, port designations, and the IP addressing scheme is a ____.

- a.) architectural floor plan b.) structural blueprint
- c.) logical topology diagram d.) physical topology diagram

____ 33. A network type that connects a few computers to each other and to the internet within a private residence is a ____.

- a.) campus area network b.) storage area network
- c.) small home network d.) metropolitan area network

____ 34. A network setup that enables computers within a home or remote office to connect back to a corporate network is called a ____.

- a.) Small Office/Home Office (SOHO)**
- b.) personal intranet**
- c.) storage area network**
- d.) global area network**

____ 35. A network infrastructure that connects many locations with hundreds or thousands of interconnected computers is a ____.

- a.) peer workgroup b.) medium to large network
- c.) small home network d.) personal area network

____ 36. A network connecting hundreds of millions of computers worldwide, such as the internet, is classified as a ____.

- a.) metropolitan network b.) campus network
- c.) worldwide network d.) local workgroup

____ 37. A network infrastructure that spans a small geographical area and provides high-speed bandwidth to internal end devices is a ____.

- a.) Storage Area Network (SAN) b.) Personal Area Network (PAN)
- c.) Local Area Network (LAN) d.) Wide Area Network (WAN)

____ 38. A network infrastructure that spans a wide geographical area to interconnect distant LANs is called a ____.

- a.) Controller Area Network (CAN)**
- b.) Wide Area Network (WAN)**
- c.) Near Field Network (NFN)**
- d.) Local Area Network (LAN)**

____ 39. Unlike a WAN which is managed by service providers, a LAN is typically administered by a single organization or ____.

- a.) public utility commission b.) individual
- c.) international committee d.) government consortium

____ 40. Between LANs and WANs, the network type that typically provides slower speed transmission links between sites is a ____.

- a.) WAN b.) SAN
- c.) SOHO d.) LAN

____ 41. The worldwide collection of interconnected LANs and WANs that is not owned by any single individual or group is called the ____.

- a.) Intranet b.) Extranet
- c.) Internet d.) Ethernet

____ 42. WAN infrastructures that interconnect LANs across the globe may use copper wires, wireless transmissions, and ____.

- a.) fiber-optic cables b.) loopback interfaces
- c.) token-ring circuits d.) serial patch bays

_____ 43. The open international standards organization responsible for developing and maintaining Internet protocols and standards is the _____.

- a.) ITU-T b.) ANSI
- c.) IETF d.) IEEE

_____ 44. The non-profit organization responsible for coordinating the allocation of global IP addresses and DNS domain names is _____.

- a.) ISO b.) ICANN
- c.) W3C d.) IETF

_____ 45. The technical body responsible for the overall architectural oversight of Internet standards and protocols is the _____.

- a.) FCC b.) EIA
- c.) NIST d.) IAB

_____ 46. A private collection of LANs and WANs internal to an organization accessible exclusively to authorized members is an _____.

- a.) extranet b.) internet
- c.) intranet d.) darknet

_____ 47. A network that provides secure, authenticated access to an organization's network for external partners, vendors, or customers is an _____.

- a.) intranet b.) extranet
- c.) ethernet d.) internet

_____ 48. An always-on, high-bandwidth Internet connection offered over television coaxial cabling by cable TV providers is called _____.

- a.) ISDN b.) dial-up
- c.) cable d.) DSL

_____ 49. A high-bandwidth, always-on Internet connection that runs over traditional copper telephone lines is called _____.

- a.) cable b.) cellular
- c.) DSL d.) satellite

_____ 50. An Internet access technology that uses a mobile phone cellular network to connect end devices to the Internet is called _____.

- a.) fiber-optic b.) cellular
- c.) powerline d.) dial-up

_____ 51. An Internet access technology that provides a major benefit to rural areas lacking terrestrial Internet Service Providers is _____.

- a.) Metro Ethernet b.) satellite
- c.) Leased Line d.) SDSL

_____ 52. An inexpensive, low-bandwidth Internet connection that operates over a standard analog telephone line using a modem is _____.

- a.) broadband cable b.) dial-up telephone
- c.) DSL d.) wireless broadband

_____ 53. A reserved, private circuit within a service provider's network connecting distant corporate offices for voice and data is a _____.

- a.) public hotspot b.) dedicated leased line

- c.) consumer DSL line d.) dial-up circuit

_____ **54.** A business-class connection technology that extends familiar LAN access technology directly into the wide area network is Metro Ethernet or _____.

- a.) Ethernet WAN b.) Frame Relay
c.) Token Ring WAN d.) ATM switch

_____ **55.** A business-grade DSL variant that provides identical bandwidth for both upload and download speeds is called _____.

- a.) VDSL b.) SDSL
c.) Dial-up d.) ADSL

_____ **56.** Before converged networks, an organization was separately cabled for data, video, and _____.

- a.) electrical power b.) security intercom
c.) HVAC controls d.) telephone

_____ **57.** A network infrastructure that delivers data, voice, and video over a single unified link using the same set of rules and standards is a _____.

- a.) dual-ring network b.) segmented network
c.) converged network d.) circuit-switched network

_____ **58.** The technologies and structural design that support the underlying infrastructure moving data across a network are called network _____.

- a.) bandwidth b.) architecture
c.) telemetry d.) firmware

_____ **59.** The four basic characteristics that underlying network architectures must address are scalability, QoS, security, and _____.

- a.) fault tolerance b.) high latency
c.) low cost d.) manual routing

_____ **60.** A network design characteristic that limits the impact of a failure by limiting the number of affected devices is called _____.

- a.) traffic shaping b.) bandwidth capping
c.) network scaling d.) fault tolerance

_____ **61.** To achieve fault tolerance and redundancy, reliable modern networks implement a _____.

- a.) circuit-switched network b.) packet-switched network
c.) centralized hub d.) token-passing bus

_____ **62.** In a packet-switched network, traffic is split into packets where each packet can theoretically take a different path to the _____.

- a.) default gateway only b.) central mainframe
c.) local broadcast domain d.) destination

_____ **63.** A network architecture that establishes dedicated physical circuits between endpoints before communication begins is a _____.

- a.) peer-to-peer network b.) converged network
c.) circuit-switched network d.) packet-switched network

_____ **64.** The characteristic of a network that allows it to expand quickly and easily to support new users and applications without degrading existing performance is _____.

- a.) confidentiality b.) fault tolerance
c.) scalability d.) latency

_____ **65.** Network designers follow accepted international standards and protocols primarily to ensure networks are _____.

- a.) scalable
- b.) circuit-switched
- c.) centralized
- d.) proprietary

_____ **66.** The primary network mechanism used to prioritize time-sensitive traffic like voice and live video to ensure reliable delivery is _____.

- a.)** Quality of Service (QoS)
- b.)** Access Control Lists (ACL)
- c.)** Network Address Translation (NAT)
- d.)** Dynamic Host Configuration (DHCP)

_____ **67.** Watching an online video that suffers from constant pauses and stutter occurs when demand exceeds bandwidth and _____ is not configured.

- a.) DNS
- b.) DHCP
- c.) QoS
- d.) SSH

_____ **68.** Network security that addresses the physical protection of routers, switches, and cabling from unauthorized access is _____.

- a.)** information security
- b.)** network infrastructure security
- c.)** cloud virtualization
- d.)** application logic

_____ **69.** Network security that focuses on protecting the actual data transmitted over the network is called _____.

- a.) infrastructure security
- b.) hardware diagnostics
- c.) information security
- d.) cable management

_____ **70.** The goal of network security that ensures only intended recipients can read data is _____.

- a.) confidentiality
- b.) scalability
- c.) integrity
- d.) availability

_____ **71.** The goal of network security that provides assurance that data has not been altered or tampered with during transmission is _____.

- a.) confidentiality
- b.) non-repudiation
- c.) availability
- d.) integrity

_____ **72.** The goal of network security that ensures authorized users have timely and dependable access to data and services is _____.

- a.) confidentiality
- b.) integrity
- c.) availability
- d.) privacy

_____ **73.** The growing trend that allows end users to use personal tools and devices to access corporate information is known as _____.

- a.)** Internet of Things (IoT)
- b.)** Network Virtualization
- c.)** Bring Your Own Device (BYOD)
- d.)** Data Center Bridging

_____ **74.** BYOD is summarized by the philosophy of allowing any device, with any ownership, to be used _____.

- a.) anonymously
- b.) anywhere
- c.) without an IP address
- d.) in secret

_____ **75.** Working with others over the network on joint projects and sharing interactive content in real time is called online _____.

- a.) benchmarking
- b.) segmentation
- c.) virtualization
- d.) collaboration

a.) Network Assistant b.) SecureCRT
c.) Packet Tracer d.) Webex Teams

a.) TelePresence b.) AnyConnect
c.) Firepower d.) Prime

a.) cloud computing b.) terminal emulation
c.) edge slicing d.) powerline networking

a.) repeater stations b.) data centers
c.) wiring racks d.) telecommunication closets

a.) private cloud b.) custom cloud
c.) hybrid cloud d.) public cloud

a.) public cloud b.) private cloud
c.) custom cloud d.) hybrid cloud

a.) partitioned cloud b.) custom cloud
c.) unified cloud d.) hybrid cloud

a.) general cloud b.) custom cloud
c.) baseline cloud d.) public cloud

a.) edge computing b.) dark fiber
c.) smart home technology d.) industrial automation

a.) optical switching b.) powerline networking
c.) cellular tethering d.) satellite broadband

a.) option b.) topology
c.) threat d.) protocol

_____ **87.** An Internet Service Provider commonly found in rural environments that connects subscribers to designated wireless access points is a _____.

- a.)** Cable Operator
- b.)** Dial-up Carrier
- c.)** Satellite Fleet
- d.)** Wireless Internet Service Provider (WISP)

_____ **88.** Wireless broadband solutions for homes and small businesses use the same cellular technology found in a _____.

- a.) rotary phone
- b.) smartphone
- c.) walkie-talkie
- d.) cable modem

_____ **89.** Malicious code and attacks such as viruses, worms, and Trojan horses originating outside the network are classified as _____.

- a.) external threats
- b.) physical hazards
- c.) system vulnerabilities
- d.) internal threats

_____ **90.** A network attack that exploits an undocumented software vulnerability on the very day it is discovered before a patch exists is a _____.

- a.) dictionary attack
- b.) zero-day attack
- c.) brute-force attack
- d.) replay attack

_____ **91.** A network attack designed to overwhelm network servers or devices with traffic to prevent legitimate users from accessing services is a _____.

- a.) privilege breach
- b.) spyware scan
- c.) phishing attack
- d.) denial of service attack

_____ **92.** Security risks originating from within an organization, such as lost or stolen devices and accidental employee mistakes, are _____.

- a.) external threats
- b.) zero-day exploits
- c.) botnet assaults
- d.) internal threats

_____ **93.** A foundational security solution on home and small office networks used to inspect and block unauthorized incoming traffic is a _____.

- a.) unmanaged hub
- b.) firewall
- c.) repeater
- d.) bridge

_____ **94.** On enterprise networks, a series of sequential permit or deny statements used on routers to filter traffic based on IP addresses is an _____.

- a.) Address Resolution Table
- b.) Access Control List (ACL)
- c.) CAM table
- d.) Dynamic Routing Table

_____ **95.** A dedicated security appliance that actively monitors network traffic to detect and block fast-spreading threats like zero-day attacks is an _____.

- a.)** Uninterruptible Power Supply
- b.)** Intrusion Prevention System (IPS)
- c.)** Network Interface Card
- d.)** Media Converter

_____ **96.** A technology that provides encrypted, secure communication channels over an untrusted public network for teleworkers is a _____.

- a.)** Wide Area Network
- b.)** Virtual Private Network (VPN)
- c.)** Local Area Network
- d.)** Dynamic Host Protocol

_____ **97.** The foundational industry certification that demonstrates knowledge of IP foundation, security, automation, and wireless is the _____.

- a.) CompTIA A+ b.) PMP
- c.) CCNA d.) CISSP

_____ **98.** Cisco's certification track at associate, specialist, and professional levels designed to validate software development and programmability skills is _____.

- a.) Routing & Switching b.) CyberOps
- c.) DevNet d.) CCIE

_____ **99.** On www.netacad.com, Networking Academy students and alumni can search for jobs with Cisco and partner employers using the _____.

- a.)** Cisco Job Matcher
- b.)** Talent Bridge Matching Engine
- c.)** Career Netlink
- d.)** Cisco Talent Scout

PART II: ANSWER KEY & STUDY REFERENCE — MODULE 1: NETWORKING TODAY

Official Solutions, Correct Terms, and Curriculum Concept References

Item #	Key	Correct Identification Term	Curriculum Topic & Reference Explanation
1	a.)	boundaries	[1.1 Networks Affect Our Lives] Networks create a 'world without boundaries' by connecting individuals and communities globally.
2	b.)	communities	[1.1 Networks Affect Our Lives] Global communities are formed by individuals sharing common interests across the global network.
3	a.)	human	[1.1 Networks Affect Our Lives] Cisco emphasizes the concept of the 'human network' that enriches human interactions.
4	d.)	end device	[1.2 Host Roles] Hosts or end devices are computers and endpoints that originate or receive data.
5	b.)	servers	[1.2 Host Roles] Servers are computers that run specialized software to provide data/services to end devices.
6	c.)	clients	[1.2 Host Roles] Clients run client software that initiates requests to servers to obtain information.
7	a.)	email server	[1.2 Host Roles] An email server runs email server software while clients use mail clients to access email.
8	b.)	browser	[1.2 Host Roles] Clients use browser software to access web pages served by a web server.
9	a.)	file server	[1.2 Host Roles] A file server stores corporate and user files which client devices access.
10	b.)	peer-to-peer network	[1.2 Peer-to-Peer] In a peer-to-peer (P2P) network, devices share resources without dedicated centralized servers.
11	b.)	lower cost	[1.2 Peer-to-Peer] Peer-to-peer advantages include: easy to set up, less complex, and

			lower cost.
12	b.)	centralized administration	[1.2 Peer-to-Peer] Peer-to-peer networks have no centralized administration, making security and backups difficult.
13	a.)	scalable	[1.2 Peer-to-Peer] P2P networks are not scalable; as more devices are added, performance and administration degrade.
14	c.)	printers	[1.2 Peer-to-Peer] P2P networks are primarily used for simple tasks like sharing files and sharing printers.
15	c.)	end device	[1.2 End Devices] An end device is where data originates from, flows through the network, and arrives.
16	a.)	intermediary network device	[1.2 Intermediary Network Devices] Intermediary devices interconnect end devices and direct network traffic.
17	c.)	wireless access points	[1.2 Intermediary Network Devices] Intermediary devices include switches, wireless access points (APs), routers, and firewalls.
18	a.)	retransmit data signals	[1.2 Intermediary Network Devices] Intermediary devices regenerate and retransmit data signals across network segments.
19	b.)	network	[1.2 Intermediary Network Devices] Intermediary devices (such as routers) maintain pathway and routing information.
20	c.)	failures	[1.2 Intermediary Network Devices] Intermediary devices notify other devices of communication errors and link failures.
21	a.)	media	[1.2 Network Media] Network media provides the physical pathway or medium through which messages travel.
22	d.)	electrical impulses	[1.2 Network Media] Copper cabling (metal wires) encodes and transmits data as electrical impulses.
23	c.)	fiber-optic cable	[1.2 Network Media] Fiber-

			optic cables use glass or plastic fibers to transmit data as pulses of light.
24	d.)	light	[1.2 Network Media] Fiber-optic cables transmit data using pulses of light.
25	b.)	electromagnetic waves	[1.2 Network Media] Wireless network media transmits signals by modulating electromagnetic wave frequencies.
26	a.)	topology diagram	[1.3 Network Representations] Topology diagrams represent network devices and interconnections using standard symbols.
27	b.)	Network Interface Card (NIC)	[1.3 Network Representations] A Network Interface Card (NIC) connects an end device to the physical network media.
28	a.)	port	[1.3 Network Representations] A physical port is an outlet/connector on a network device where cables attach.
29	c.)	interfaces	[1.3 Network Representations] Interfaces are specialized ports on a network device connecting to specific networks.
30	c.)	port	[1.3 Network Representations] Cisco notes that often the terms 'port' and 'interface' are used interchangeably.
31	a.)	physical topology diagram	[1.3 Topology Diagrams] Physical topology diagrams show physical locations (rooms, racks) and cable routes.
32	c.)	logical topology diagram	[1.3 Topology Diagrams] Logical topology diagrams illustrate devices, ports, and the addressing scheme of the network.
33	c.)	small home network	[1.4 Networks of Many Sizes] Small home networks connect a few computers to each other and to the Internet.
34	a.)	Small Office/Home Office (SOHO)	[1.4 Networks of Many Sizes] A SOHO network enables a computer within a home or

			remote office to connect to a corporate network.
35	b.)	medium to large network	[1.4 Networks of Many Sizes] Medium to large networks have many locations with hundreds or thousands of computers.
36	c.)	worldwide network	[1.4 Networks of Many Sizes] Worldwide networks connect hundreds of millions of computers globally.
37	c.)	Local Area Network (LAN)	[1.4 LANs and WANs] A LAN is a network infrastructure that spans a small geographical area.
38	b.)	Wide Area Network (WAN)	[1.4 LANs and WANs] A WAN spans a wide geographical area and interconnects LANs across cities or regions.
39	b.)	individual	[1.4 LANs and WANs] LANs are administered by a single organization or individual.
40	a.)	WAN	[1.4 LANs and WANs] WANs typically provide slower speed links between interconnected LANs.
41	c.)	Internet	[1.4 The Internet] The Internet is a worldwide collection of interconnected LANs and WANs.
42	a.)	fiber-optic cables	[1.4 The Internet] WANs may use copper wires, fiber optic cables, and wireless transmissions.
43	c.)	IETF	[1.4 The Internet] The Internet Engineering Task Force (IETF) develops and promotes Internet standards.
44	b.)	ICANN	[1.4 The Internet] ICANN coordinates IP address allocation and Domain Name System management.
45	d.)	IAB	[1.4 The Internet] The Internet Architecture Board (IAB) helps maintain structure and architecture on the Internet.
46	c.)	intranet	[1.4 Intranets and Extranets] An intranet is a private internal network meant only for organization members.
47	b.)	extranet	[1.4 Intranets and Extranets] An extranet provides secure

			access to external individuals who work for different organizations.
48	c.)	cable	[1.5 Internet Connections] Cable provides high bandwidth, always-on internet offered by cable television providers.
49	c.)	DSL	[1.5 Internet Connections] DSL (Digital Subscriber Line) is a high-bandwidth, always-on connection over telephone lines.
50	b.)	cellular	[1.5 Internet Connections] Cellular internet connects devices using a cell phone network.
51	b.)	satellite	[1.5 Internet Connections] Satellite internet provides a major benefit to rural areas without standard ISPs.
52	b.)	dial-up telephone	[1.5 Internet Connections] Dial-up telephone is an inexpensive, low-bandwidth option using a modem and phone line.
53	b.)	dedicated leased line	[1.5 Businesses Internet Connections] Dedicated leased lines are reserved circuits connecting distant offices with private networking.
54	a.)	Ethernet WAN	[1.5 Businesses Internet Connections] Ethernet WAN (Metro Ethernet) extends LAN access technology into the WAN.
55	b.)	SDSL	[1.5 Businesses Internet Connections] Symmetric Digital Subscriber Line (SDSL) provides symmetric bandwidth for businesses.
56	d.)	telephone	[1.5 The Converging Network] Traditionally, organizations had separate cabling for telephone, video, and data networks.
57	c.)	converged network	[1.5 The Converging Network] Converged networks carry multiple services (data, voice, video) on one link using unified standards.
58	b.)	architecture	[1.6 Reliable Networks] Network Architecture refers to

			the technologies that support the infrastructure moving data.
59	a.)	fault tolerance	[1.6 Reliable Networks] The 4 basic requirements of a reliable network are Fault Tolerance, Scalability, QoS, and Security.
60	d.)	fault tolerance	[1.6 Fault Tolerance] A fault tolerant network limits the impact of a failure by requiring multiple redundant paths.
61	b.)	packet-switched network	[1.6 Fault Tolerance] Reliable networks provide redundancy by implementing a packet-switched network.
62	d.)	destination	[1.6 Fault Tolerance] Packet switching splits traffic into packets that can each take a different path to the destination.
63	c.)	circuit-switched network	[1.6 Fault Tolerance] Circuit-switched networks establish dedicated circuits, unlike packet-switched networks.
64	c.)	scalability	[1.6 Scalability] A scalable network can expand quickly and easily to support new users without impacting performance.
65	a.)	scalable	[1.6 Scalability] Standards and protocols enable network equipment from diverse vendors to scale seamlessly.
66	a.)	Quality of Service (QoS)	[1.6 Quality of Service] Quality of Service (QoS) is the primary mechanism used to ensure reliable delivery of content.
67	c.)	QoS	[1.6 Quality of Service] Breaks and pauses in streaming video happen when demand exceeds bandwidth and QoS isn't configured.
68	b.)	network infrastructure security	[1.6 Network Security] Network infrastructure security involves physical security and preventing unauthorized device access.
69	c.)	information security	[1.6 Network Security] Information security protects the information or data transmitted over the network.

70	a.)	confidentiality	[1.6 Network Security] Confidentiality ensures that only intended recipients can read the data.
71	d.)	integrity	[1.6 Network Security] Integrity provides assurance that the data has not been altered during transmission.
72	c.)	availability	[1.6 Network Security] Availability provides assurance of timely and reliable access to data for authorized users.
73	c.)	Bring Your Own Device (BYOD)	[1.7 Bring Your Own Device] Bring Your Own Device (BYOD) allows users to use their personal devices for work and communication.
74	b.)	anywhere	[1.7 Bring Your Own Device] BYOD means 'any device, with any ownership, used anywhere'.
75	d.)	collaboration	[1.7 Online Collaboration] Online collaboration allows users to work with others over the network on joint projects.
76	d.)	Webex Teams	[1.7 Online Collaboration] Cisco Webex Teams is a multifunctional collaboration tool for messaging, images, videos, and links.
77	a.)	TelePresence	[1.7 Video Communication] Cisco TelePresence powers interactive video collaboration where everyone can meet from anywhere.
78	a.)	cloud computing	[1.7 Cloud Computing] Cloud computing allows us to store personal files or backup data on servers over the Internet.
79	b.)	data centers	[1.7 Cloud Computing] Cloud computing is made possible by data centers from which companies lease server/storage services.
80	d.)	public cloud	[1.7 Cloud Computing] Public clouds are available to the general public through a pay-per-use model or for free.
81	b.)	private cloud	[1.7 Cloud Computing] Private clouds are intended for a specific organization or entity such as the government.

82	d.)	hybrid cloud	[1.7 Cloud Computing] Hybrid clouds are composed of two or more cloud types (e.g., part custom and part public).
83	b.)	custom cloud	[1.7 Cloud Computing] Custom clouds are built to meet the needs of a specific industry, such as healthcare or media.
84	c.)	smart home technology	[1.7 Technology Trends in the Home] Smart home technology integrates technology into everyday appliances to interconnect them.
85	b.)	powerline networking	[1.7 Powerline Networking] Powerline networking allows devices to connect to a LAN using existing electrical outlets.
86	a.)	option	[1.7 Powerline Networking] Powerline networking connects devices where network cables or wireless are not viable options.
87	d.)	Wireless Internet Service Provider (WISP)	[1.7 Wireless Broadband] A WISP connects subscribers to designated access points or hotspots in rural environments.
88	b.)	smartphone	[1.7 Wireless Broadband] Wireless broadband uses the same cellular technology used by a smartphone.
89	a.)	external threats	[1.8 Security Threats] External threats originate outside an organization and include viruses, worms, and Trojans.
90	b.)	zero-day attack	[1.8 Security Threats] Zero-day attacks exploit software vulnerabilities on day one of discovery before patches exist.
91	d.)	denial of service attack	[1.8 Security Threats] Denial of Service (DoS) attacks disable services by swamping them with excessive traffic.
92	d.)	internal threats	[1.8 Security Threats] Internal threats include lost/stolen devices, accidental misuse by employees, and malicious staff.
93	b.)	firewall	[1.8 Security Solutions] Firewall filtering blocks unauthorized access to the

			network.
94	b.)	Access Control List (ACL)	[1.8 Security Solutions] Access Control Lists (ACL) filter network traffic based on IP addresses and services.
95	b.)	Intrusion Prevention System (IPS)	[1.8 Security Solutions] An Intrusion Prevention System (IPS) actively identifies and mitigates fast-spreading attacks.
96	b.)	Virtual Private Network (VPN)	[1.8 Security Solutions] Virtual Private Networks (VPN) provide secure remote access for employees over the Internet.
97	c.)	CCNA	[1.9 The IT Professional] The Cisco Certified Network Associate (CCNA) certification validates foundational networking knowledge.
98	c.)	DevNet	[1.9 The IT Professional] DevNet certifications validate software development and network programmability skills.
99	b.)	Talent Bridge Matching Engine	[1.9 The IT Professional] The Talent Bridge Matching Engine connects students and alumni with job opportunities.